Midterm 2017, "The Mathematical Foundations of Political Methodology"

You have 120 minutes for this exam. You may use a calculator and refer to your Pemberton & Rau book. Please do not use the internet during the exam. Email your solutions, *with your work shown*, to Milena. Solutions without work will get no credit.

1. Given the following hold, what do we know about the relationship between sets $A$ and $C$?

   (a) $A \cap B = B$ and $B \cap C \neq \emptyset$

   (b) $A \cup B = B$ and $B \cap C \neq \emptyset$

   (c) $B \cap D = C$ and $B \cap C = A$

2. Find the limit of each of the following sequences as $n \to \infty$, if the limit exists.

   (a)
   $$\frac{4n^2 - 1}{(n+1)^2}$$

   (b)
   $$\frac{(-1)^n n^{\frac{1}{2}}}{7 + n^{\frac{3}{2}}}$$

   (c)
   $$\frac{(-1)^n n^{\frac{1}{2}}}{n^{\frac{1}{4}} + n^{\frac{1}{2}}}$$

3. Use the $(\delta, \epsilon)$ definition of a limit to prove that the function
   $$f(x) = \frac{2}{3}x - 17$$
   is everywhere continuous.

4. Let
   $$f(x) = \frac{x}{x^2 + 1} + 5$$

   (a) Find the critical points of $f$ and characterize them as maxima, minima, or inflection points.

   (b) Find the regions on which $f$ is convex and concave.

   (c) Find any asymptotes that $f$ may have.

5. Perform the following matrix operations:

   (a) Find the inverse of
$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

   (b) Find the rank of
$$\begin{bmatrix} 1 & 2 & 5 \\ -3 & 2 & -1 \\ 0 & 4 & 7 \\ 2 & 0 & 3 \end{bmatrix}$$

   (c) Check whether the following matrix is positive definite, positive semidefinite, negative definite, negative semidefinite, or none of the above.
$$\begin{bmatrix} 1 & 3 & 2 \\ 3 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix}$$

6. Find and classify all of the critical points of the following functions:

   (a)
$$f(x, y) = -10x^2 + 4xy - \frac{2y^2}{5}$$

   (b)
$$f(x, y, z) = -x^2 + \log(x) - \frac{1}{y} - y - z^2 + z \quad \text{for } x, y, z > 0$$

   (c)
$$f(x, y) = x^2 + y^2 - xy \text{ subject to the constraint that } 2x - y = 3$$